Calculus III: Multivariable Integration & Stokes' Theorem

Problem 1: Flux through a Paraboloid

Compute the outward flux of the vector field F through the surface S bounded by the paraboloid.

Gauss Divergence Theorem: $$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) \, dV$$

Problem 2: Circulation along Closed Path

Stokes' Theorem: $$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

Table 1: Test Vector Fields and Boundary Curves

Field Index	Vector Field F(x, y, z)	Boundary Condition	Domain
Field A	F = (y, -x, z^2)	C: x^2 + y^2 = 1, z = 0	Unit disk in xy-plane
Field B	F = (2z, 3x, 5y)	S: x^2 + y^2 + z^2 = 4, z >= 0	Upper hemisphere
Field C	F = (e^y, e^z, e^x)	Tetrahedron at origin	x+y+z <= 1, x,y,z >= 0